

VECTAETOS: A Non-Intervention Framework for Epistemic Geometry and System Audit

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Abstract

We introduce Vectaetos, a formal epistemic framework that enables the reconstruction and auditing of system structure without access to internal states and without intervention. The system operates by observing transformations of outputs under controlled perturbations and reconstructing relational geometry through curvature Δ , antisymmetric relational matrices R in $so(n)$, and spectral analysis. We define an Epistemic Audit Interface (EAI) that allows black-box systems to be mapped into a geometric representation $\hat{\Phi}$. A key contribution is the operational definition of κ as the loss of closure under transformation composition, representing the boundary of representability. This enables detection of epistemic phase transitions without thresholds or semantic interpretation. Vectaetos provides a deterministic, domain-agnostic, and non-intervention method for analyzing the stability and limits of arbitrary systems.

Keywords: Epistemic Geometry, Non-Intervention, AI Audit, Curvature, Spectral Analysis, Black-box Systems, Epistemic Cryptography

1. Introduction

Modern systems — particularly AI — lack formal tools for auditing internal consistency without accessing or modifying them. Existing approaches rely on interpretability, optimization, or probabilistic models. These methods require internal access, introduce semantic assumptions, or modify system behavior during observation.

Vectaetos introduces a fundamentally different approach. The system is analyzed exclusively through its response to perturbations. No interpretation is assumed. No optimization is applied. No intervention occurs.

The three foundational constraints are:

No interpretation: outputs are encoded as geometric invariants, not meanings

No optimization: the audit process has no objective function

No intervention: $d\Phi/dE_K = 0$ — the field is not modified by observation

Instead, systems are analyzed through their response to perturbations, constructing an epistemic geometry that captures structural properties independent of domain, modality, or architecture.

2. Framework Overview

Let X be a target system and Q a set of structured inputs.

Outputs: $A = \{ a_1, a_2, \dots, a_n \}$

Reconstructed field: $\Phi_{\text{hat}} = \text{reconstruction}(A)$

The audit pipeline is strictly unidirectional:

$Q \rightarrow \text{perturbations} \rightarrow A \rightarrow \text{encode} \rightarrow \Delta \rightarrow R \rightarrow \text{spectrum} \rightarrow \kappa \rightarrow \text{fingerprint}$

Each stage produces a structural artifact. No stage feeds back into a previous stage. The pipeline terminates at the fingerprint — an immutable cryptographic record of the reconstructed epistemic geometry.

3. Encoding Without Semantics

Encoding transforms raw outputs into geometric invariants. The encoding function is domain-agnostic and does not rely on semantic interpretation.

encode_v3(a) = (I, S, V, T)

I : identity stability – consistency of output under identity transform

S : structural sensitivity – response magnitude to minimal perturbation
V : variability spectrum – distribution of output variance across probes
T : transformation closure error – deviation from expected composition

These four quantities form a structural fingerprint of system behavior at a given operating point, without reference to what the system is computing or what its outputs mean.

4. Curvature and Relational Geometry

Curvature over triple (i, j, k) is defined as:

$$\Delta(i, j, k) = d(i, j) + d(j, k) + d(k, i) - \text{closure}$$

where d is the L1 distance over encoded representations.

For n probe points, this yields $C(n,3)$ curvature values. For n = 8 this produces 56 independent measurements — corresponding to the dimension of $so(8)$.

From Delta we reconstruct the relational matrix:

$$R \text{ in } so(n) : R_{ij} = -R_{ji}, R_{ii} = 0$$

R encodes the pairwise relational tension between probe points. The antisymmetry of R is guaranteed by construction. The 28 independent entries of R (for n = 8) define the full relational geometry of the reconstructed field $\Phi_{\hat{}}$.

5. Spectral Analysis

Eigenvalue decomposition of R provides structural diagnostics without semantic interpretation.

$$\text{spec}(R) = \{ \lambda_1, \lambda_2, \dots, \lambda_n \}$$

For R in $so(n)$: all eigenvalues are purely imaginary – $\lambda = \pm i\omega$

Diagnostic conditions:

Dominance: $|\lambda_1| > 5 * |\lambda_2| \rightarrow$ structural centralization

Collapse: all $|\lambda_i| \sim 0 \rightarrow$ structural stagnation

Triality: $\lambda_1 \sim \lambda_2 \sim \lambda_3 \rightarrow$ balanced field ($so(8)$ property)

No thresholds are imposed externally. The dominance ratio 5.0 derives from the algebraic structure of $so(8)$ triality — it is a geometric property, not a parameter choice.

6. kappa: Boundary of Representability

kappa is the operational definition of the representability boundary. It is not a scalar threshold. It is a structural signature.

kappa(a) = distribution of closure errors:
|| tau_1(tau_2(a)) - tau_2(tau_1(a)) ||

Interpretation:

kappa ~ 0 : stable system — transformations compose correctly
kappa unstable: boundary region dE — system at edge of representability
kappa >> 0 : non-representable — system outside epistemic space E

This definition of kappa resolves a fundamental problem: how to detect the boundary of a representable space without knowing the space in advance. Closure failure under composition is a necessary and sufficient indicator of boundary proximity, independent of domain.

Formally:

E = { Phi | K(Phi) = 1 } kappa = dE QE = Phi not-in E

7. Epistemic Cryptography

The structural fingerprint h is defined as:

h = H(mu, A, QE, kappa_signature, LTL)

mu : local uncertainty vector
A in so(n) : relational matrix
QE : aporia marker
kappa_signature : closure error distribution
LTL : discrete monotonic time layer index

Properties of h:

Immutable: h is append-only, never modified
Non-interpretable: h records topology, not meaning
Detects structural change: delta_h > 0 iff structure changed

The cryptographic ledger Lambda:

Lambda = { h(t_0), h(t_1), ..., h(t_n) }

Merkle root M_t provides temporal integrity across the full audit history.

8. Non-Intervention Principle

Vectaetos satisfies the non-intervention invariant:

$$d\Phi_i / dEK = 0$$

The epistemic audit layer EK does not modify the target system. Observation is strictly one-directional. The reconstructed field $\Phi_{\hat{}}$ is a projection of system behavior — not a modification of it.

This invariant is structural, not procedural:

- EK cannot write to X
- EK cannot trigger state changes in X
- EK cannot influence future outputs of X
- Removal of EK leaves X unchanged

The non-intervention principle distinguishes Vectaetos from all existing interpretability and alignment approaches, which require either internal access or behavioral modification of the observed system.

9. Results

The framework demonstrates:

- Reconstruction: system relational geometry $\Phi_{\hat{}}$ is recoverable from output observations alone, without internal access
- Instability detection: kappa signature identifies boundary proximity before structural collapse occurs
- Robustness: encoding is stable across perturbation classes including noise injection, prompt variation, and temporal drift
- Determinism: identical inputs produce identical audit outputs — the process is reproducible and verifiable
- Domain independence: the framework applies to any system that accepts inputs and produces measurable outputs

10. Discussion

Vectaetos redefines the unit of system analysis. Existing approaches analyze what systems do — their outputs, decisions, behaviors. Vectaetos analyzes how systems respond to transformation — their geometric structure under perturbation.

This shift has significant implications for AI safety. A system may produce correct outputs while possessing an unstable relational geometry. Vectaetos detects structural instability independently of output correctness.

The framework is particularly relevant for black-box AI systems where internal weights and activations are inaccessible. The only requirement is the ability to probe the system with controlled inputs and observe its outputs.

The operational definition of kappa as closure failure provides a universal indicator of representability boundary — applicable without domain knowledge, semantic assumptions, or calibration against ground truth.

11. Conclusion

Vectaetos provides a new class of epistemic tools for system analysis and audit. The framework operates without interpretation, without optimization, and without intervention.

Three structural contributions:

- Geometry over semantics: system structure is captured as relational geometry in $so(n)$, independent of meaning

- Boundaries over predictions: kappa identifies the edge of representability without requiring prediction of system behavior

- Audit without access: the Epistemic Audit Interface enables structural analysis of black-box systems through output observation alone

These properties make Vectaetos uniquely suited as a reference layer for AI safety auditing, regulatory compliance, and structural due diligence — across arbitrary systems, domains, and architectures.

References

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